



Successful cryptanalysis on RSA type modulus $N = p^2q$

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ABSTRACT

As internet technology advances and our interactions increasingly take place online, cryptography emerges as a valuable tool to address security concerns. Cryptography serves as a means to guarantee the protection of privacy and confidential information, thereby instilling confidence when sharing and exchanging such data with other parties. One of the benefits of cryptography is providing confidentiality, which protects our information; either data in transit or data at ease. Three new attacks have been proposed on RSA type modulus $N = p^2q$. The equation $eX - NY = Z - (p^2k + q^2m)Y$ is the basis for the first attack involves random values of k and m such that k being a multiple of 2 and m being a multiple of 3, both being integers with $|p^2k + q^2m| < N^{1/2}$ and $\gcd(X, Y) = 1$. If $Z < \frac{|p^2k - q^2m|}{3(p^2k + q^2m)^2} N^{1/3} Y$ and $XY < \frac{N}{4(p^2k + q^2m)^2}$, then by using continued fractions, factoring N can be accomplished within polynomial time. This paper also suggested the vulnerabilities t RSA cryptosystem moduli $N_s = p_s^2q_s$ for $t \geq 2$ and $s = 1, \dots, t$ for the second and third attack. The attacks are effective if there exists a relationship between t RSA public keys (N_t, e_t) expressed as $e_sx - N_sy = z_s - (p_s^2k + q_s^2m)y_s$ or $e_sx_s - N_sy = z_s - (p_s^2k + q_s^2m)y$, where the variables x, x_s, y, y_s , and z_s are sufficiently small. The importance lies in the presence of an algorithm that operates within probabilistic polynomial time, which is capable of accepting public parameters as input and yield the factors p and q as output. By employing this algorithm, one can verify whether a key falls within the vulnerable class or not. This characteristic can be valuable when designing a cryptosystem during the key generation process, as it helps prevent the inadvertent creation of weak keys.

1. Introduction

With the advancement of internet technology, the realm of cryptography has become a valuable tool, particularly when it comes to matters of security. Cryptography offers a means to ensure the safeguarding of our privacy and confidential information, instilling confidence when sharing and exchanging such information with others. One key advantage of cryptography is its ability to provide confidentiality by protecting our data, whether it is in transit or at rest. It is essential for both the sender and the receiver to possess the necessary key in order to carry out their respective operations successfully. Thus, irrespective of how complex the encryption and decryption algorithms may be, the security of the key itself is paramount to avoid potential problems.

The first workable asymmetric cryptosystem was introduced by Rivest, Shamir, and Adleman (RSA) in 1978. In the RSA cryptosystem, a modulus N is generated by multiplying two prime numbers, p and q , of the same bit-size. Simultaneously, a public exponent e , is chosen as a positive integer that is coprime with a parameter $\phi(N) = (p-1)(q-1)$ and a private exponent d , is used for decryption, satisfying the Diophantine equation $ed - k\phi(N) = 1$ [12].

Initially conceived in the RSA cryptosystem, the inclusion of a small private exponent d aimed to minimize the computational costs associated with decryption. As a result, the overall quantity of modular multiplications needed for modular exponentiation as well as the cost of decryption decreased. While RSA still maintains a satisfactory level of security when combined with suitable cryptographic techniques, extensive literature exists exploring its cryptanalysis. Since its inception, the RSA system has become the subject of widespread research in the field of cryptology. For instance, a significant discovery demonstrates that if the decryption exponent d is smaller than $\frac{1}{3}N^{\frac{1}{4}}$, the RSA cryptosystem becomes insecure when utilizing continued fractions [15]. Subsequently, Boneh and Durfee adjusted this threshold to $d < N^{0.292}$ using Coppersmith's method to detect small solutions of modular univariate polynomials [3].

As part of these endeavors to ensure acceptable levels of security while maximizing computational efficiency, numerous variations of RSA have been suggested. Among these variants are those based on moduli in the form of $N = p^2q$, which have found widespread use in cryptography.

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In the pursuit of enhanced throughput, numerous variant of RSA have been presented to enable the transmission of large data sets and achieve computational effectiveness while preserving the desired level of security. These innovative variations utilize the concept of "Prime Power RSA," which involves expressing the modulus N in the form $N = p^r q$, where $r \geq 2$. To enhance the efficiency of the decryption process, Takagi [14] introduced a variant that employs a modulus of the form $N = p^{b-1}q$, where $b \geq 3$. The public and private exponents are defined as inverses modulo $\lambda'(N) = \text{lcm}(p-1)(q-1)$. Consequently, the private exponent d satisfies the congruence relation $ed \equiv 1 \pmod{\lambda'(N)}$.

Following that, Okamoto [11] introduced a public key cryptosystem that offers provable security comparable to factoring moduli in the form $N = p^2q$. Taking a different approach, Boneh and Durfee [4] conducted a cryptanalysis of Prime Power RSA when a partial knowledge of p is available. According to May [8], due to the reduced amount of confidential information stored in the prime factors for a fixed bit-size of N in comparison to RSA, it is anticipated that cryptosystems employing moduli of the form $N = p^r q$, where $r > 1$, are more susceptible to attacks on the secret key compared to the original RSA scheme. Consequently, these schemes are more vulnerable to attacks like Wiener attack and Boneh-Durfee attack that exploit small secret keys. Consequently, they devised two attacks on moduli $N = p^r q$ with $r \geq 2$, which improved the constraints on the size of d by utilizing Coppersmith's method to solve modular univariate polynomial equations.

In addition, Sarkar proved that factorization of $N = p^2q$ can be accomplished by using lattice reduction techniques if $d < N^{0.395}$ [13]. Later on, Asbullah and Ariffin [1] showed that if the term $N - (2N^{2/3} - N^{1/3})$ is a good approximation of $\phi(N)$ satisfy the key equation $ed - k\phi(N) = 1$, then one can factor $N = p^2q$ in polynomial time. Recently, Asbullah et al. [2] demonstrated that factorization of $N = p^2q$ can be accomplished within polynomial time if e satisfy the generalized key equation together with the condition on the parameters. This is the reason why we focus on the variant of RSA modulus $N = p^2q$ in our work.

Our contribution. This paper introduces three novel attacks targeting the RSA-type modulus $N = p^2q$. With k being a multiple of 2 and m being a multiple of 3, the first attack is based on the equation $eX - NY = Z - (p^2k + q^2m)Y$. The attack is successful under the conditions $|p^2k - q^2m| < N^{1/2}$ and $\gcd(X, Y) = 1$. By utilizing continued fractions, the paper demonstrates that if $Z < \frac{|p^2k - q^2m|}{3(p^2k + q^2m)} N^{1/3} Y$ and $XY < \frac{N}{4(p^2k + q^2m)^2}$, the factorization of N is possible within polynomial time. Next, the weaknesses of the t RSA cryptosystem moduli $N_s = p_s^2 q_s$ are considered for the second and third attacks where $t \geq 2$ and $s = 1, \dots, t$. These attacks exploit the relationship between k RSA public keys (N_s, e_s) described by $e_s x - N_s y_s = z_s - (p_s^2 k + q_s^2 m) y_s$ or $e_s x_s - N_s y = z_s - (p_s^2 k + q_s^2 m) y$, with the parameters x, x_s, y, y_s , and z_s chosen to be sufficiently small. The significance lies in having a probabilistic polynomial time algorithm that takes public parameters as input and produces the factors p and q as output. By employing this algorithm, one can verify whether a key falls within the vulnerable class or not. This characteristic can be valuable when designing a cryptosystem during the key generation process, as it helps prevent the inadvertent creation of weak keys.

The structure of this article is as outlined below. We commence by providing a concise overview of the continuous fractions expansion, lattice basis reduction, and simultaneous Diophantine approximation techniques, which are discussed in Section 2. The cryptanalysis of the first attack using continued fractions is presented in Section 3. Following that, Section 4 covers the second and third attacks. Section 5 we summarize our findings and puts forth suggestions for future research.

2. Preliminaries

2.1. Continued fraction

Here, we define the terms continuous fraction and simultaneous

Diophantine approximation via lattice reduction, which is going to be employed all through the rest of the study. The basic approximation problems follows; given a real number α and a positive integer N , find the rational number with denominator q less than or equal to N that is nearest to α .

By examining the convergent in the continued fractions expansion method, it becomes possible to achieve this approximation. Continued fractions find numerous applications in number theory and cryptographic scenarios. We utilize continued fractions to obtain good Diophantine approximations for both rational and irrational numbers, which aids in solving Diophantine equations and subsequently enables the development of attacks on certain instances of RSA. We present the fundamental properties of continued fractions expansion as follows [9].

Definition 2.1. (Continued Fractions Expansion). A continued fractions is an expression of the form

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \dots}}} = [a_0, a_1, \dots, a_n, \dots],$$

which, for simplicity, it is possible to express this as $x = [a_0, a_1, \dots, a_n, \dots]$. In the case where $x \in \mathbb{Q}$, the calculation of the continued fractions expansion would terminate at a finite index n , resulting in $x = [a_0, a_1, \dots, a_n]$.

The following theorem describe Legendre's Theorem [5]. The Legendre's Theorem is the main result from the theory of the continued fraction expansion which many people use to attack RSA.

Theorem 2.2. (*Legendre's Theorem*). Let x be a real positive number. If X and Y are positive integers such that $\gcd(X, Y) = 1$ and

$$|x - \frac{Y}{X}| < \frac{1}{2X^2}$$

then $\frac{Y}{X}$ is one convergent of the continued fraction expansion of x .

2.2. Simultaneous Diophantine approximations

Here, we begin with an introduction some of the basic concepts that are related to lattice and LLL algorithm.

Let's assume that u_1, u_2 , and u_3 are d linearly independent vectors for $\mathbb{R}^n$ with $d \leq n$. The set of all integer linear combinations of the vectors $u_1, \dots, u_d$ that traverse the lattice have the following form.

$$\mathcal{L} = \left\{ \sum_{i=1}^d x_i u_i \mid x_i \in \mathbb{Z} \right\}.$$

The set $(u_1, \dots, u_d)$ is known as the basis of $\mathcal{L}$ and its dimension is d . The determinant of $\mathcal{L}$ is defined as $\det(\mathcal{L}) = \sqrt{\det(U^T U)}$ where U is the matrix of the u_i 's in the canonical basis of $\mathbb{R}^n$. $\|v\|$ is defined to be the Euclidean norm of a vector $v \in \mathcal{L}$. The main goal in lattice reduction is to find a short non-zero vector in $\mathcal{L}$ [10]. Hence, the LLL algorithm that is introduced by [6] produced a reduced basis and the sizes of the reduced basis vectors is fixed by the following theorem [7].

Theorem 2.3. [6]. Let $\mathcal{L}$ be a lattice of dimension τ with a basis $\{w_1, \dots, w_\tau\}$. The LLL algorithm produces a reduced basis $\{b_1, \dots, b_\tau\}$ satisfying

$$\|b_1\| \leq \|b_2\| \leq \dots \leq \|b_\tau\| \leq 2^{\frac{\tau(\tau-1)}{2(\tau+1)}} \det(\mathcal{L})^{\frac{1}{\tau+1}},$$

for all $1 \leq i \leq \tau$.

The LLL algorithm is widely recognized for its effectiveness in solving the problem of simultaneous Diophantine approximations. According to Dirichlet's classical theorem, there are integers $p_1, \dots, p_n$ and a positive integer $q \leq \varepsilon^{-n}$ that satisfy the inequality $|q\alpha_i - p_i| < \varepsilon$ for $1 \leq i \leq n$. The LLL algorithm presents a technique for discovering

simultaneous Diophantine approximations for rational numbers by employing a lattice that incorporates real numbers [10].

Theorem 2.4. (Simultaneous Diophantine Approximations). [10]. Given rational numbers $\alpha_1, \dots, \alpha_n$ and a value $0 < \varepsilon < 1$, there exists a polynomial time algorithm that is capable of computing integers $p_1, \dots, p_n$ and a positive integer q such that

$$\max_i |q\alpha_i - p_i| < \varepsilon \quad \text{and} \quad q \leq 2^{n(n-3)/4} \cdot 3^n \cdot \varepsilon^{-n}. \quad (1)$$

3. Cryptanalysis on $eX - NY = Z - (p^2k + q^2m)Y$ using continued fraction

This section is dedicated to the presentation of our latest cryptanalysis according to the generalized key equation $eX - NY = Z - (p^2k + q^2m)Y$. Consider u being an integer multiple of 2 and v being an integer multiple of 3, together with some restriction on some parameters using continued fraction expansion. We prove that even though k and m are not coprime we still can obtain information through one of the convergents of the continued fractions expansion of $\frac{e}{N}$ which will lead to the factorization of $N = p^2q$ in polynomial time. We begin this section with the following lemmas.

Lemma 3.1. Suppose that $N = p^2q$ with $q < p < 2q$. Let $k, m \in \mathbb{N}$ and $|p^2k + q^2m| < N^{1/2}$. Let S be an approximation of $p^2k + q^2m$ such that $|p^2k + q^2m - S| < \frac{|p^2k - q^2m|}{3(p^2k + q^2m)} N^{1/3}$, then $kmq = \left\lfloor \frac{S^2}{4N} \right\rfloor$.

Proof. Suppose that $S = p^2k + q^2m + x$ where $|x| < \frac{|p^2k - q^2m|}{3(p^2k + q^2m)} N^{1/3}$. Determine

$$\begin{aligned} (p^2k - q^2m)^2 &= (p^2k - q^2m)(p^2k - q^2m) \\ &= (p^2k)^2 - (p^2q^2km) - (p^2q^2km) + (q^2m)^2 \\ &= (p^2k)^2 - 2(p^2q^2km) + (q^2m)^2 \\ &= (p^2k)^2 + 2(p^2q^2km) - 2(p^2q^2km) - 2(p^2q^2km) + (q^2m)^2 \\ &= (p^2k + q^2m)^2 - 4(p^2q^2km) \\ &= (p^2k + q^2m)^2 - 4Nqkm \end{aligned}$$

Hence we get

$$(p^2k - q^2m)^2 = (p^2k + q^2m)^2 - 4Nqkm \quad (2)$$

and consider

$$\begin{aligned} S^2 - 4Nqkm &= (p^2k + q^2m + x)^2 - 4Nqkm \\ &= (p^2k + q^2m + x)(p^2k + q^2m + x) - 4Nqkm \\ &= (p^2k)^2 + 2(p^2q^2km) + (q^2v)^2 + 2xp^2k + 2xq^2m + x^2 - 4Nqkm \\ &= (p^2k + q^2m)^2 + 2x(p^2k + q^2m) + x^2 - 4Nqkm \end{aligned}$$

By using (2), the equation can be reformulated as

$$S^2 - 4Nqkm = (p^2k + q^2m)^2 + 2|x|(p^2k + q^2m) + x^2 \quad (3)$$

Since $|p^2k - q^2m| < N^{1/2}$ and $|x| < \frac{|p^2k - q^2m|}{3(p^2k + q^2m)} N^{1/3} < N^{1/3}$, as a result there are

$$\begin{aligned} |S^2 - 4Nqkm| &= (p^2k + q^2m)^2 + 2|x|(p^2k + q^2m) + x^2 \\ &< (N^{1/2})^2 + 2(p^2k + q^2m) \frac{|p^2k - q^2m|}{3(p^2k + q^2m)} N^{1/3} + (N^{1/3})^2 \\ &< N + \frac{2}{3}(N^{1/2})N^{1/3} + N^{2/3} \\ &< 2N \end{aligned}$$

Hence, we obtain $|S^2 - 4Nqkm| < 2N$. The result of dividing by $4N$ is that

$$\left| \frac{S^2}{4N} - kmq \right| < \frac{2N}{4N} = \frac{1}{2}$$

Thus, we have $kmq = \left\lfloor \frac{S^2}{4N} \right\rfloor$ which terminates the proof. $\square$

Now, we introduce useful lemma which showing the relation of X and Y with $\frac{e}{N}$.

Lemma 3.2. Suppose that $N = p^2q$ with $q < p < 2q$. Let e be an exponent satisfying an equation $eX - (N - (p^2k + q^2m))Y = Z$. Consider k being an integer multiple of 2 and m being an integer multiple of 3 and $\gcd(X, Y) = 1$. If $XY < \frac{N}{4(p^2k + q^2m)}$ and $|Z| < \frac{|p^2k - q^2m|}{3(p^2k + q^2m)} N^{1/3}Y$, then $\frac{Y}{X}$ is a convergent of the continued fraction $\frac{e}{N}$.

Proof. Consider an exponent e that fulfill a given equation

$$eX - (N - (p^2k + q^2m))Y = Z \quad (4)$$

with $|Z| < \frac{|p^2k - q^2m|}{3(p^2k + q^2m)} N^{1/3}Y$. Thus, $|Z| < (p^2k + q^2m)Y$. Then, if we divide (4) by NX , we get

$$\begin{aligned} \left| \frac{e}{N} - \frac{Y}{X} \right| &= \frac{|eX - NY|}{NX} \\ &= \frac{|Z - (p^2k + q^2m)Y|}{NX} \\ &\leq \frac{|Z| + |p^2k + q^2m|Y}{NX} \\ &\leq \frac{2|p^2k + q^2m|Y}{NX} \end{aligned}$$

To make use of the Legendre Theorem, it is essential to observe that

$$\begin{aligned} \left| \frac{e}{N} - \frac{Y}{X} \right| &< \frac{1}{2X^2} \\ \frac{2|p^2k + q^2m|Y}{NX} &< \frac{1}{2X^2} \\ XY &< \frac{N}{4|p^2k + q^2m|} \end{aligned}$$

Hence, if $XY < \frac{N}{4|p^2k + q^2m|}$, then $\frac{2|p^2k + q^2m|Y}{NX} < \frac{1}{2X^2}$. Therefore, $\frac{Y}{X}$ is convergent of continued fraction $\frac{e}{N}$. $\square$

Next theorem proves the way to the factorization of $N = p^2q$.

Theorem 3.3. Suppose that $N = p^2q$ with $q < p < 2q$. Let e be an exponent satisfies an equation $eX - (N - (p^2k + q^2m))Y = Z$. Consider k being an integer multiple of 2 and m being an integer multiple of 3. Let $|p^2k + q^2m| < N^{1/2}$ and $\gcd(X, Y) = 1$. If $|Z| < \frac{|p^2k - q^2m|}{3(p^2k + q^2m)} N^{1/3}Y$ and $XY < \frac{N}{4|p^2k + q^2m|}$, then it is possible to factorize N in polynomial time.

Proof. Suppose we have an exponent e that satisfies the equation $eX - (N - (p^2k + q^2m))Y = Z$. Let $XY < \frac{N}{4|p^2k + q^2m|}$ and $|Z| < \frac{|p^2k - q^2m|}{3(p^2k + q^2m)} N^{1/3}Y$ as $\gcd(X, Y) = 1$ satisfy Lemma 3.2. Therefore, $\frac{Y}{X}$ is one of the convergent of the continued fraction $\frac{e}{N}$. Then, we reorder (4) as

$$(p^2k + q^2m)Y - NY + eX = Z \quad (5)$$

Divide (5) by Y , we get

$$(p^2k + q^2m) - N + \frac{eX}{Y} = \frac{Z}{Y} \quad (6)$$

Now, let $S = N - \frac{eX}{Y}$. We can rewritten (6) as $(p^2k + q^2m) - S = \frac{Z}{Y}$. Thus, we obtain

$$|(p^2k + q^2m) - S| = \frac{|Z|}{Y} < \frac{|p^2k - q^2m|}{3(p^2k + q^2m)} N^{1/3}Y.$$

By Lemma 3.1, if $|p^2k - q^2m| < N^{1/2}$ and $|(p^2k + q^2m) - S| < \frac{|p^2k - q^2m|}{3(p^2k + q^2m)} N^{1/3}$, then $kmq = \left\lceil \frac{S^2}{4N} \right\rceil$. Thus, we get $q = \gcd\left(\left\lceil \frac{S^2}{4N} \right\rceil, N\right)$. $\square$

3.1. Numerical example of the first cryptanalysis

Example 1. As an illustration of our proposed attack, we consider N and e as follows.

$$\begin{aligned} N &= 101792191315163 \\ e &= 82074376638383 \end{aligned}$$

Let say N and e fulfill all the conditions in Theorem 3.3. Then, we compute the continued fraction of $\frac{e}{N}$ which gives the list of the convergent as follows

$$\left[0, 1, \frac{4}{5}, \frac{25}{31}, \frac{154}{191}, \frac{333}{413}, \frac{820}{1017}, \frac{1153}{1430}, \frac{1973}{2447}, \dots\right].$$

Based on the convergent, we can skip the first and the second entry and begin with the convergent $\frac{4}{5}$. Hence, we have

$$S = N - \frac{eX}{Y} = -\frac{32031179312631}{4}$$

with

$$\left\lceil \frac{S^2}{4N} \right\rceil = 1574894330.$$

If we compute $\gcd(1574894330, 101792191315163) = 1$, then, we try for next convergent $\frac{25}{31}$, we get

$$S = N - \frac{eX}{Y} = \frac{499107089171}{25}$$

and

$$\left\lceil \frac{S^2}{4N} \right\rceil = 978888.$$

Thus, if we compute $\gcd(978888, 101792191315163) = 40787$, then it is possible to factor N since $q = 40787$ and $p = \sqrt{\frac{N}{q}} = 49957$.

4. System of equation $eX - NY = Z - (p^2k + q^2m)Y$

In this section, we introduce the system of equation satisfying $e_sx - N_sy_s = z_s - (p_s^2k + q_s^2m)y_s$ and $e_sx_s - N_sy = z_s - (p_s^2k + q_s^2m)y$. To obtain the parameters (x, y_s) or (y, x_s) , we employ lattice basis reduction following the transformation of the equations into a simultaneous Diophantine approximation. Concurrently, we determine the prime factors p_s and q_s of each $N_s = p_s^2q_s$ in polynomial time.

4.1. The attack on t moduli $N_s = p_s^2q_s$ via generalized key equation $e_sx - N_sy_s = z_s - (p_s^2k + q_s^2m)y_s$

Next, we initiate the attack on a set of t moduli, each following the form $N_s = p_s^2q_s$, where all moduli share the same size N , with $N = \min N_i$. We look at the moduli that satisfy k equations of the form $e_sx - N_sy_s = z_s - (p_s^2k + q_s^2m)y_s$. By appropriately choosing small values for the unknown parameters x, y_s , and z_s , it becomes possible to factor all k moduli in the form $N_s = p_s^2q_s$. This leads us to the following theorem, which relies on Lemma 3.1 and Theorem 3.3.

Theorem 4.1. Suppose that $t \geq 2$, let $N_s = p_s^2q_s$, $1 \leq s \leq t$ be t moduli. Assume $N = \min N_s$. Let e_s , $s = 1, \dots, t$ be t public exponents. Define $\delta = \frac{t-3\lambda}{3(1+t)}$. Consider k being an integer multiple of 2 and m being an integer

multiple of 3. If there is an integer $x < N^\delta$ and t integers $y_s < N^\delta$ for $s = 1, \dots, t$, along with the condition $|z_s| < \frac{|p_s^2k - q_s^2m|}{3(p_s^2k + q_s^2m)} N^{1/3} y_s$, such that $e_sx - N_sy_s = z_s - (p_s^2k + q_s^2m)y_s$, then it becomes feasible to factor t moduli $N_1, \dots, N_t$ in polynomial time.

Proof. Consider $t \geq 2$ and the equation $e_sx - (N_s - (p_s^2k + q_s^2m))y_s = z_s$ which can be written as $e_sx - N_sy_s = z_s - (p_s^2k + q_s^2m)y_s$ with $s = 1, \dots, t$. Hence,

$$\left| \frac{e_s}{N_s}x - y_s \right| = \frac{|z_s - y_s(p_s^2k + q_s^2m)|}{N_s} \quad (7)$$

Consider $N = \min N_s$ and assume that $y_s < N^\delta$ and $|z_s| < \frac{|p_s^2k - q_s^2m|}{3(p_s^2k + q_s^2m)} N^{1/3} y_s$.

Then, $|z_s| < y_s N^{1/3} < N^{\delta+1/3}$. Set $p_s^2k + q_s^2m < N^{\delta+1/3}$, we have

$$\begin{aligned} \left| \frac{z_s - y_s(p_s^2k + q_s^2m)}{N_s} \right| &\leq \frac{|z_s| + y_s(p_s^2k + q_s^2m)}{N} \\ &< \frac{N^{\delta+1/3} + N^\delta(N^{\delta+1/3})}{N} \\ &< \frac{2N^{\delta+1/3}}{N} \\ &= 2N^{\delta-1/3} \end{aligned} \quad (8)$$

Through the application of Theorem 2.4, we substitute (8) in (7). As a result we have

$$\left| \frac{e_s}{N_s}x - y_s \right| < 2N^{\delta-1/3}$$

The connection between $\left| \frac{e_s}{N_s}x - y_s \right| < 2N^{\delta-1/3}$ and $|qa_i - p_i| < \varepsilon$ satisfies the condition stated in Theorem 2.4, can be observed.

Now, the presence of an integer x and the integers y_s has been demonstrated which $\varepsilon = 2N^{\delta-1/3}$, $\delta = \frac{t-3\lambda}{3(1+t)}$. We get

$$N^\delta \cdot \varepsilon^t = 2^t N^{\delta+t\delta-1/3+t} = 2^t$$

Since $2^t < 2^{\frac{t(t-3)}{4}} \cdot 3^t$ for $t \geq 2$, we have $N^\delta \cdot \varepsilon^t < 2^{\frac{t(t-3)}{4}} \cdot 3^t$ by applying Theorem 2.4. It follows that if $x < N^\delta$, then $x < 2^{\frac{t(t-3)}{4}} \cdot 3^t \cdot \varepsilon^{-t}$. In brief for $s = 1, \dots, t$, we obtain

$$\left| \frac{e_s}{N_s}x - y_s \right| < \varepsilon, \quad x < 2^{\frac{t(t-3)}{4}} \cdot 3^t \cdot \varepsilon^{-t}$$

By using the LLL algorithm leads us to find x and y_s which the conditions of Theorem 2.4 are met. Subsequently, by utilizing the equation $e_sx - (N_s - (p_s^2k + q_s^2m))y_s = z_s$, we obtain

$$(p_s^2k + q_s^2m) - N_s + \frac{e_sx}{y_s} = \frac{z_s}{y_s}.$$

If $|z_s| < \frac{|p_s^2k - q_s^2m|}{3(p_s^2k + q_s^2m)} N^{1/3} y_s$, then $\frac{z_s}{y_s} < \frac{|p_s^2k - q_s^2m|}{3(p_s^2k + q_s^2m)} N^{1/3}$ and $S_s = N_s - \frac{e_sx}{y_s}$ is an approximation of $p_s^2k + q_s^2m$.

Therefore, based on Lemma 3.1 and Theorem 3.3, this indicates that $kmq = \left\lceil \frac{S_s^2}{4N} \right\rceil$ for $S_s = N_s - \frac{e_sx}{y_s}$, $s = 1, \dots, t$, we compute $q_s = \gcd\left(\left\lceil \frac{S_s^2}{4N} \right\rceil, N_s\right)$. Hence, it is feasible to factorize t moduli $N_s = p_s^2q_s$. This terminates the proof. $\square$

4.1.1. Numerical example of the second cryptanalysis

Example 2. As an illustration of our proposed attack, we consider three moduli and three public exponents as follows

$$\begin{aligned}
N_1 &= 46933608167335710028982913397, \\
N_2 &= 33026309601240645591521912101, \\
N_3 &= 30524619907256967141971123339, \\
e_1 &= 23923356631563315920844198053, \\
e_2 &= 28657070116945893614872993033, \\
e_3 &= 12233602495076595447872448467.
\end{aligned}$$

Then, $N = \min(N_1, N_2, N_3) = 30524619907256967141971123339$. For $t = 3$, we have $\delta = \frac{t-3t}{3(1+t)} = \frac{1}{10}$ and $\varepsilon = 2N^{\delta-\frac{1}{3}+\lambda} \approx 0.2246682246607288$. Set $k = 4$ and $m = 6$. Suppose that we consider the parameter C as defined in [10], (Appendix A, page 196) and consider $n = t = 3$, leads to

$$C = \left[3^{n+1} \cdot 2^{\frac{(n+1)(n-4)}{4}} \cdot e^{-n-1} \right] = 15896.$$

We look at the lattice $\mathcal{L}$ spanned by the rows of the matrix

$$M = \begin{bmatrix} 1 & -[Ce_1/N_1] & -[Ce_2/N_2] & -[Ce_3/N_3] \\ 0 & C & 0 & 0 \\ 0 & 0 & C & 0 \\ 0 & 0 & 0 & C \end{bmatrix}.$$

Then, the LLL algorithm is applied to lattice $\mathcal{L}$ leads to the reduced basis together with the matrix as follows

$$K = \begin{bmatrix} 257 & 162 & 7 & 198 \\ 212 & -856 & 748 & 720 \\ 1550 & -260 & 970 & -2084 \\ -1405 & 1774 & 1941 & 402 \end{bmatrix}.$$

Now, we obtain

$$K \cdot M^{-1} = \begin{bmatrix} 257 & 131 & 223 & 103 \\ 212 & 108 & 184 & 85 \\ 1550 & 790 & 1345 & 621 \\ -1405 & -716 & -1219 & -563 \end{bmatrix}.$$

According to the first row of the above matrix, we obtain $x = 257, y_1 = 131, y_2 = 223$ and $y_3 = 103$.

By applying x and y_s for $s = 1, 2, 3$, we look at the relation $S_s = N_s - \frac{e_s x}{y_s}$ is an approximation of $p_s^2 k + q_s^2 m$.

Hence, by using Lemma 3.1 and Theorem 3.3, this implies that $kmq = \left\lceil \frac{S_s^2}{4N_s} \right\rceil$ for $S_s = N_s - \frac{e_s x}{y_s}$. Then, we compute

$$\left\lceil \frac{S_1^2}{4N_1} \right\rceil = 75626689272, \left\lceil \frac{S_2^2}{4N_2} \right\rceil = 67266758904, \left\lceil \frac{S_3^2}{4N_3} \right\rceil = 65523477576.$$

For each $s = 1, 2, 3$, we find $q_s = \gcd\left(\left\lceil \frac{S_s^2}{4N_s} \right\rceil, N_s\right)$ and we obtain

$$q_1 = 3151112053, \quad q_2 = 2802781621, \quad q_3 = 2730144899.$$

It is possible to factor three moduli N_1, N_2 and N_3 since

$$p_1 = 3859313407, \quad p_2 = 3432696391, \quad p_3 = 3343738331.$$

4.2. The attack on t Moduli $N_s = p_s^2 q_s$ via generalized key equation $e_s x_s - N_s y = z_s - (p_s^2 k + q_s^2 m)y$

Now, we proceed with the attack on t moduli of the form $N_s = p_s^2 q_s$ where all N_s have the same size, $N = \max N_s$. We examine a set of moduli that satisfy t equations $e_s x_s - N_s y = z_s - (p_s^2 k + q_s^2 m)y$. Through the theorem presented, it has been demonstrated that moduli of the form $N_s = p_s^2 q_s$ can be factored when the unknown parameters x, y_s , and z_s are sufficiently small. Our objective is to combine this knowledge with the application of the LLL algorithm utilizing Lemma 3.1 and Theorem 3.3.

Theorem 4.2. Consider $t \geq 2$ and $N_s = p_s^2 q_s$. For $1 \leq s \leq t$, t moduli have the same size N , where $N = \max N_s$ and t public exponents satisfying $\min e_s = N^\gamma$. Define $\delta = \frac{k(3\gamma-2-3\lambda)}{3t+1}$. Suppose k being an integer multiple of 2 and m being an integer multiple of 3 and $\gcd(X, Y) = 1$. If there exist an integer $y < N^\delta$, t integers $x_s < N^\delta$ and $|z_s| < \frac{|p_s^2 k - q_s^2 m|}{3(p_s^2 k + q_s^2 m)} N^{1/3} y$ such that $e_s x_s - N_s y = z_s - (p_s^2 k + q_s^2 m)y$ then it is feasible to factor t moduli $N_1, \dots, N_t$ in polynomial time.

Proof. Consider that $t \geq 2$ and the equation $e_s x_s - (N_s - (p_s^2 k + q_s^2 m))y = z_s$ which can be written as $e_s x_s - N_s y = z_s - (p_s^2 k + q_s^2 m)y$. Hence,

$$\left| \frac{N_s}{e_s} y - x_s \right| = \frac{|z_s - (p_s^2 k + q_s^2 m)y|}{e_s} \quad (9)$$

Consider $N = \max N_s$. Assume $y < N^\delta$ and $|z_s| < \frac{|p_s^2 k - q_s^2 m|}{3(p_s^2 k + q_s^2 m)} N^{1/3} y$ and later $|z_s| < y N^{1/3} < N^{\delta+1/3}$. Consider as well that $\min_s e_s = N^\gamma$. Define $p_s^2 k + q_s^2 m < N^{\frac{2}{3}+\alpha}$ with $0 < \alpha < 1/3$, then

$$\begin{aligned}
\frac{|z_s - (p_s^2 k + q_s^2 m)y|}{e_s} &\leq \frac{|z_s| + y(p_s^2 k + q_s^2 m)}{N^\gamma} \\
&< \frac{N^{\delta+1/3} + N^\delta \left(N^{\frac{2}{3}+\lambda} \right)}{N^\gamma} \\
&< \frac{2N^{\delta+\frac{2}{3}+\lambda}}{N^\gamma} \\
&= 2N^{\delta+\frac{2}{3}+\lambda-\gamma}
\end{aligned} \quad (10)$$

Upon applying Theorem 2.4, we make a substitution of (10) into (9). Next, we achieve

$$\left| \frac{N_s}{e_s} y - x_s \right| < 2N^{\delta+\frac{2}{3}+\lambda-\gamma}.$$

We can see the relation between $\left| \frac{N_s}{e_s} y - x_s \right| < 2N^{\delta+\frac{2}{3}+\lambda-\gamma}$ and $|q\alpha_i - p_i| < \varepsilon$ satisfy the condition stated in Theorem 2.4.

Now, the presence of an integer y and the integers x_s have been demonstrated with $\varepsilon = 2N^{\delta+\frac{2}{3}+\lambda-\gamma}$ and $\delta = \frac{t(3\gamma-2-3\lambda)}{3(t+1)}$. Then, we obtain

$$N^\delta \cdot \varepsilon^t = N^\delta \left(2N^{\delta+\frac{2}{3}+\lambda-\gamma} \right)^t = 2^t \left(N^{\delta+\delta t+\frac{2}{3}t+\lambda t-\gamma t} \right) = 2^t.$$

Since $2^t < 2^{\frac{t(t-3)}{4} \cdot 3^t}$ for $t \geq 2$, we have $N^\delta \cdot \varepsilon^t < 2^{\frac{t(t-3)}{4} \cdot 3^t}$ by applying Theorem 2.4. Consequently, if $y < N^\delta$, then $y < 2^{\frac{t(t-3)}{4} \cdot 3^t} \cdot \varepsilon^{-t}$. In brief for $s = 1, \dots, t$, we have

$$\left| \frac{N_s}{e_s} y - x_s \right| < \varepsilon, \quad y < 2^{\frac{t(t-3)}{4} \cdot 3^t} \cdot \varepsilon^{-t}.$$

By using the LLL algorithm allows us to find y and x_s for $s = 1, \dots, t$ when the conditions of Theorem 2.4 are met. Subsequently, by employing the equation $e_s x_s - (N_s - (p_s^2 k + q_s^2 m))y = z_s$, we obtain

$$(p_s^2 k + q_s^2 m) - N_s + \frac{e_s x_s}{y} = \frac{z_s}{y}$$

If $|z_s| < \frac{|p_s^2 k - q_s^2 m|}{3(p_s^2 k + q_s^2 m)} N^{1/3} y$, then $\frac{|z_s|}{y} < \frac{|p_s^2 k - q_s^2 m|}{3(p_s^2 k + q_s^2 m)} N^{1/3}$ and $S_s = N_s - \frac{e_s x_s}{y}$ is an approximation of $p_s^2 k + q_s^2 m$.

Therefore, based on Lemma 3.1 and Theorem 3.3, this indicates that $kmq = \left\lceil \frac{S_s^2}{4N_s} \right\rceil$ for $S_s = N_s - \frac{e_s x_s}{y}$, $s = 1, \dots, t$, we compute $q_s = \gcd\left(\left\lceil \frac{S_s^2}{4N_s} \right\rceil, N_s\right)$. Hence, it is feasible to factorize s moduli $N_s = p_s^2 q_s$. This terminates the proof. $\square$

4.2.1. Numerical example of the third cryptanalysis

Example 3. As an illustration of our proposed attack, we consider the following three moduli and three public exponents

$$\begin{aligned} N_1 &= 31251979063989640981837620337, \\ N_2 &= 44547484903769389005287075947, \\ N_3 &= 40479311005823331075635360531, \\ e_1 &= 25070268737245259447485269305, \\ e_2 &= 42233329571077130026922263619, \\ e_3 &= 31774082620664711982267697253. \end{aligned}$$

Then, $N = \max(N_1, N_2, N_3) = 44547484903769389005287075947$. We also obtain $\min(e_1, e_2, e_3) = N^\delta$ with $\delta \approx 0.9912853592$. If $t = 3$, then we have $\delta = \frac{t(3\gamma-2-3\lambda)}{3(t+1)} = 0.09346401950$ and $\epsilon = 2N^{\delta+\frac{2}{3}+\lambda-\gamma} \approx 0.2561446423645776$. Set $k = 10$ and $m = 15$.

Suppose that we consider the parameter C as defined in [10], (Appendix A, page 196) and consider $n = t = 3$, leads to

$$C = \left[3^{n+1} \cdot 2^{\frac{(n+1)(n-4)}{4}} \cdot e^{-n-1} \right] = 9408.$$

We look at the lattice $\mathcal{L}$ spanned by the rows of the matrix

$$M = \begin{bmatrix} 1 & -[CN_1/e_1] & -[CN_2/e_2] & -[CN_3/e_3] \\ 0 & C & 0 & 0 \\ 0 & 0 & C & 0 \\ 0 & 0 & 0 & C \end{bmatrix}.$$

Then, the LLL algorithm is applied to lattice $\mathcal{L}$ leads to the reduced basis together with the matrix as follows

$$K = \begin{bmatrix} 73 & -57 & -37 & -39 \\ 767 & -561 & 131 & -879 \\ -661 & -645 & 1727 & 549 \\ 1153 & -1935 & -1091 & 1647 \end{bmatrix}.$$

Now, we obtain

$$K \cdot M^{-1} = \begin{bmatrix} 73 & 91 & 77 & 93 \\ 767 & 956 & 809 & 977 \\ -661 & -824 & -697 & -842 \\ 1153 & 1437 & 1216 & 1469 \end{bmatrix}.$$

According to the first row of the above matrix, we obtain $y = 73$, $x_1 = 91$, $x_2 = 77$ and $x_3 = 93$.

By applying y and x_s for $s = 1, 2, 3$, we look at the relation $S_s = N_s - \frac{e_s x_s}{y}$ is an approximation of $p_s^2 k + q_s^2 m$.

Hence, by using Lemma 3.1 and Theorem 3.3, this implies that $kmq = \left\lceil \frac{S_s^2}{4N} \right\rceil$ for $S_s = N_s - \frac{e_s x_s}{y}$. Then, we compute

$$\left\lceil \frac{S_1^2}{4N_1} \right\rceil = 412749604950, \left\lceil \frac{S_2^2}{4N_2} \right\rceil = 464517168450, \left\lceil \frac{S_3^2}{4N_3} \right\rceil = 449923267650.$$

Table 1

Comparing Our Findings with Prior Research.

Reference	Variant Key Equations	Enable conditions
[13]	$ed = 1 + k(N - p^2 - pq + p)$	$d < N^{0.395}$
[1]	$ed - k\phi(N) = 1$	$2p^{5/3} p^{1/3} - q^{1/3} < \frac{1}{3} N^\beta, d < N^{\frac{1-\beta}{2}}$
[2]	$er - (Ns + t) = ap^2 + \beta q^2$	$1 \leq s < r < \frac{N}{3(ap^2 + \beta q^2)}, t < \frac{ ap^2 - \beta q^2 }{3(ap^2 + \beta q^2)} N^{1/3}$
Our Finding [Theorem 3.3]	$eX - NY = Z - (p^2 k + q^2 m)Y$	$Z < \frac{ p^2 k - q^2 m }{3(p^2 k + q^2 m)} N^{1/3} Y$ and $XY < \frac{N}{4(p^2 k + q^2 m)}$
Our Finding [Theorem 4.1]	$e_s x - N_s y = z_s - (p_s^2 k + q_s^2 m)y_s$	$x, y_s < N^\delta, \delta = \frac{t-3\lambda t}{3(1+t)}, z_s < \frac{ p_s^2 k - q_s^2 m }{3(p_s^2 k + q_s^2 m)} N^{1/3} y_s$
Our Finding [Theorem 4.2]	$e_s x_s - N_s y = z_s - (p_s^2 k + q_s^2 m)y$	$y, x_s < N^\delta, \delta = \frac{k(3\gamma-2-3\lambda)}{3t+1}, z_s < \frac{ p_s^2 k - q_s^2 m }{3(p_s^2 k + q_s^2 m)} N^{1/3} y$

For each $s = 1, 2, 3$, we find $q_s = \gcd\left(\left\lceil \frac{S_s^2}{4N_s} \right\rceil, N_s\right)$ and we obtain

$$q_1 = 2751664033, \quad q_2 = 3096781123, \quad q_3 = 2999488451.$$

It is possible to factor three moduli N_1, N_2 and N_3 since

$$p_1 = 3370086583, \quad p_2 = 3792768533, \quad p_3 = 3673609241.$$

4.2.2. Comparative analysis

In this section, we examine our results in relation to earlier discoveries in security analysis associated with $N = p^2 q$, specifically focusing on the structure of the adapted key equations and their corresponding conditions. The contrasts are depicted in Table 1.

Referring to Table 1 and the provided references [1,2,13], it is proved that the previous research can be categorized as a form of cryptanalysis resembling the RSA key equation $ed - k\phi(N) = 1$ and a magnified generalized Diophantine equation. The previous findings necessitated the establishment of conditions for key pairs (e, N) and their associated generalized parameters. These attacks typically involve the integration of the continued fraction method and lattice reduction techniques to develop a novel approach for factoring N .

As a result, our approach allows for the factorization of $N = p^2 q$, targeting a set of vulnerable keys with specific criteria outlined in Theorems 3.3., 4.1, and 4.2, respectively. The conditions placed on our parameters distinguish themselves from those imposed by previous results. The importance of incorporating our results as an additional

INPUT: The public key modulus (N, e) satisfy $N = p^2 q$ and Theorem 3.3.

OUTPUT: The prime factor p, q .

1. Compute the continued fraction $\frac{e}{N}$.
2. Calculate $S = N - \frac{eX}{Y}$ for each convergent $\frac{Y}{X}$ of $\frac{e}{N}$.
3. Obtain value of $\left\lceil \frac{S^2}{4N} \right\rceil$.
4. Determine $q = \gcd\left(\left\lceil \frac{S^2}{4N} \right\rceil, N\right)$.
5. Deduce $p = \sqrt{\frac{N}{q}}$ if $1 < q < N$.

Algorithm 3.1.

measure in the key generation process to ensure that crypto-designers or implementors do not accidentally generate weak keys.

5. Conclusion and future work

In conclusion, Three new attacks have been proposed on RSA type modulus $N = p^2q$. The equation $eX - NY = Z - (p^2k + q^2m)Y$ provides the basis for the first attack in such a way k being an integer multiple of 2 and m being an integer multiple of 3 together with $|p^2k - q^2m| < N^{1/2}$ and $\gcd(X, Y) = 1$. If $Z < \frac{|p^2k - q^2m|}{3(p^2k + q^2m)} N^{1/3} Y$ and $XY < \frac{N}{4|p^2k + q^2m|}$, then by using continued fractions, factorization of N can be accomplished within polynomial time. We have successfully proved the significance lies in having in the presence of an algorithm that operates within probabilistic polynomial time, which is capable of accepting public parameters as input and yield the factors p and q as output. By employing this algorithm, one can verify whether a key falls within the vulnerable class or not. This characteristic can be valuable when developing a cryptographic system during the key generation process, as it helps prevent the inadvertent creation of weak keys. Subsequently, our attention turned to the system of generalized key equations, which take the form of $e_s x - N_s y_s = z_s - (p_s^2 k + q_s^2 m) y_s$ for the second attack and $e_s x_s - N_s y = z_s - (p_s^2 k + q_s^2 m) y$ for the third attack. We successfully demonstrated the effectiveness of these three attacks when the parameters x, x_s, y, y_s , and z_s are suitably small. Moreover, we also established that both of these attacks allow for the simultaneous factorization of k RSA moduli in the form $N_s = p_s^2 q_s$ using the LLL algorithm.

There are numerous additional possibilities for further exploration regarding the variant being discussed. Future research could potentially propose more generalized key equations that highlight efficient ways to recover the prime factors p and q in polynomial time. Moreover, it is valuable to discover alternative methods for solving the factorization problem using cryptanalysis techniques other than the ones focused on in this article, such as continued fraction expansion and lattice basis reduction. It would be an impressive achievement if the encryption and decryption time could be reduced by utilizing a small private exponent. Additionally, the security of the RSA variant under partial key exposure attacks can be analyzed when the prime factors p and q share a significant number of least significant bits (LSBs) or most significant bits (MSBs).

CRedit authorship contribution statement

Normahirah Nek Abd Rahman: Writing – original draft, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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